LAB#1 GUIDE

Fast-track Your Wireless Sensing Application Development Using MCXW71 and Low-Power Motion Wakeup Sensors

Hands-on Lab#1 Guide

Contents

Hands-On Lab#1 Overview:

Tamper detection using low-power motion wakeup sensors:
Securing Your Assets with MCXW71 Wireless MCU

Brief Description

This example uses **FRDM-MCXW71** and on-board **FXLS8974CF** accelerometer sensor to demonstrate autonomous detection of tampering/theft/abuse on device using low-power motion wakeup feature and transmit ALERT message via BLE wireless UART.

Details

The core idea is to leverage advanced sensor technology to protect high-value or secure assets. For example, smart meters can monitor for tampering, enhancing home security by detecting unauthorized access to safes or lockers. Personal medical devices can be safeguarded against misuse, and laptops or tablets can be protected from theft. In industrial settings, warehouse theft detection and machine tampering alerts can significantly reduce losses and improve security. This technology can benefit even simple applications like door open/close detection.

Solution

This example demonstrates ease-of-enablement using NXP's FRDM-MCXW71 and sensors development ecosystem to accelerate prototyping for your multiple such applications.

The **Lab#1** solution comprises of:

- **MCX W71** for transmitting tamper detection alerts via BLE.
- 3-axis accelerometer **FXLS8974CF** for low-power motion wakeup.

Recommended Sensor

<u>FXLS8974CF: 3-Axis Accelerometer:</u>

The **FXLS8974CF** is a **3-axis accelerometer** targeted for application requiring **low-power motion wake up**. This sensor has **SDCD (sensor data change detection) embedded block which implements an efficient and flexible inertial event detection function** to detect various inertial events like no-motion/motion detecting tamper/theft/abuse on an asset. This ultra-low power wake-up on motion can trigger host MCU to wake-up or go back to deep sleep mode when no motion detected autonomously.

Key Features:

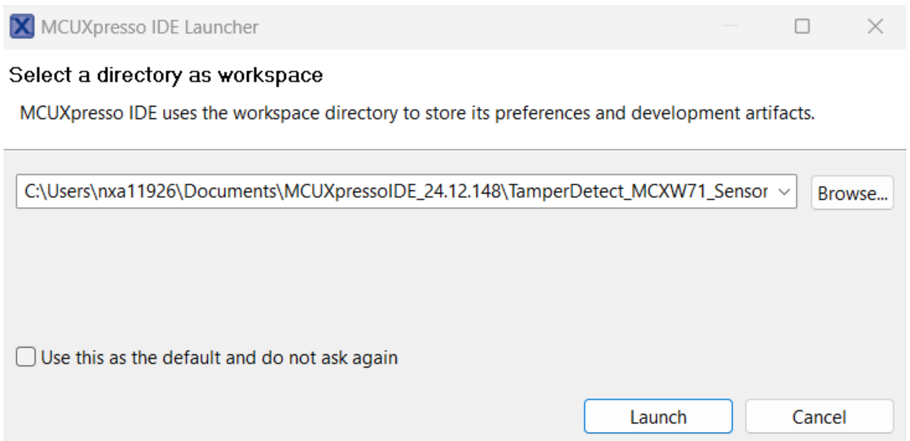
- **Ultra Low power**: < 1 µA in low power wakeup mode (0.78 Hz to 6.25 Hz ODR)
- **Sensor Data Change Detection (SDCD) function**: highly configurable digital window comparator for easy / efficient implementation of low-power motion detection
- **Self-Test Diagnostic**: Can be run in field to assess device health (unaffected by device orientation or motion)
- **±2 to ±16 g** (user selectable) full scale range
- **12-bit** Sensor Data Output Resolution
- **I²C / SPI** (pin configurable) digital interfaces
- **2 x 2 mm DFN Package**, 0.4 mm pitch with wettable flanks
- **-40 to 105 °C** operating temperature
- **EMC Class III** Compliant

Target Applications:

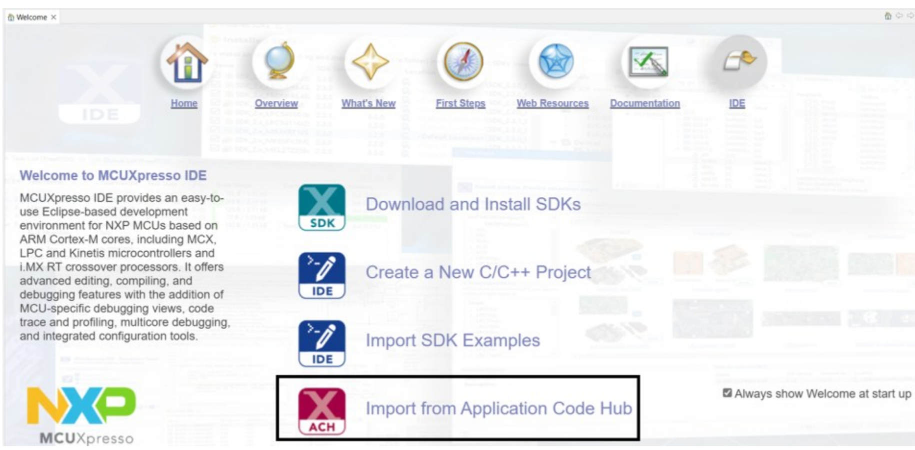
- Asset tracking/Inventory management
- Patient activity monitoring
- Metering anti-tampering
- Drug delivery.
- Medical wearables
- Home Security
- Equipment monitoring
- Vibration Sensing
- Camera Stabilization

Get Hands-on Workshop Lab Examples from ACH

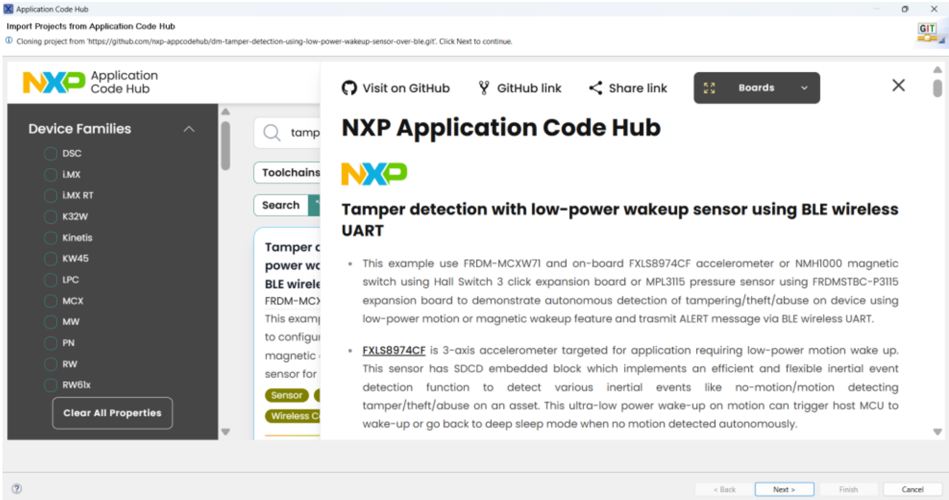
1) Launch MCUXpresso IDE and create workspace name as "TamperDetect_MCXW71_Sensors".



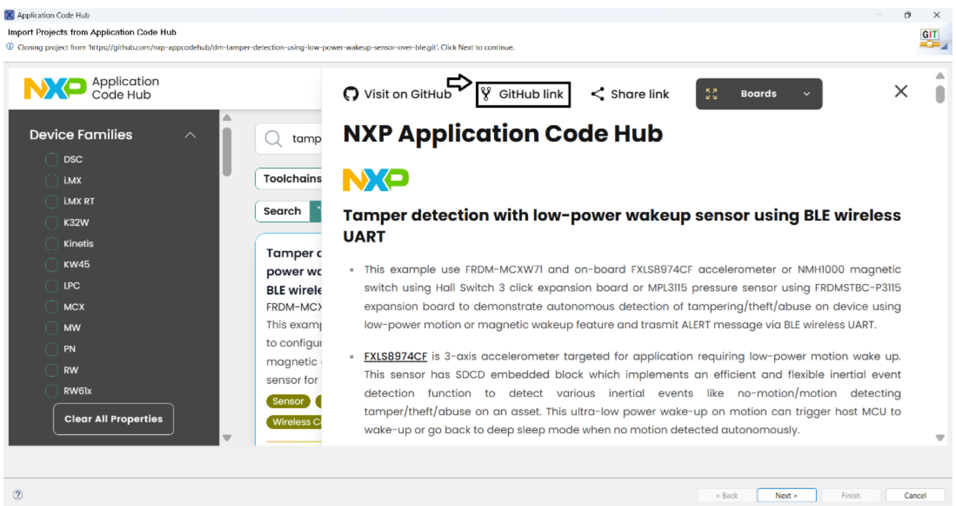
2) Select "Import from Application Code Hub" on the welcome screen.



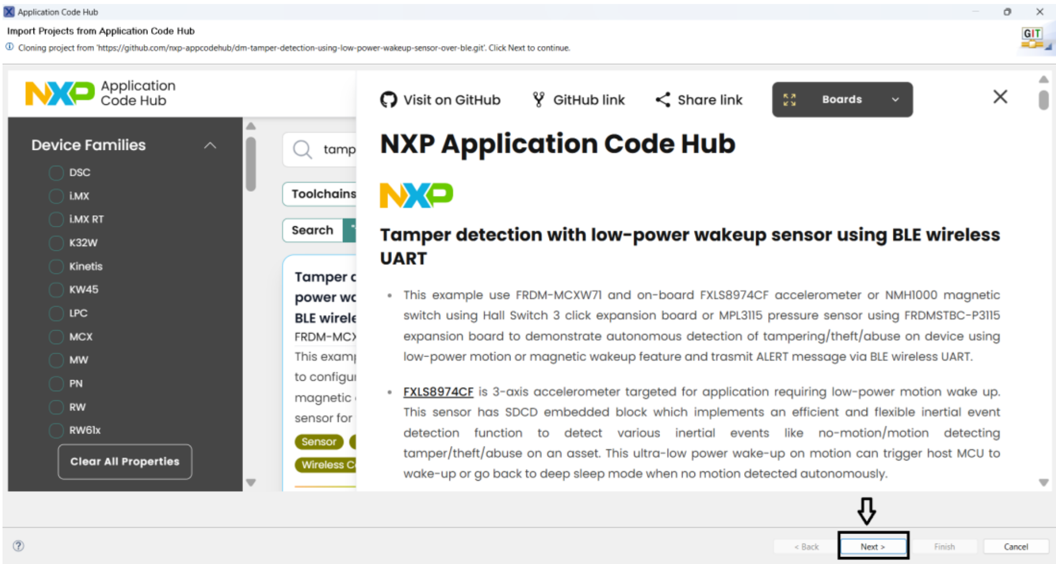
3) It will open a view, type in "Tamper detection FRDM-MCXW" in the search window. Select the "**Tamper Detection with low-power sensors using BLE wireless UART**" example as shown in the snapshot.



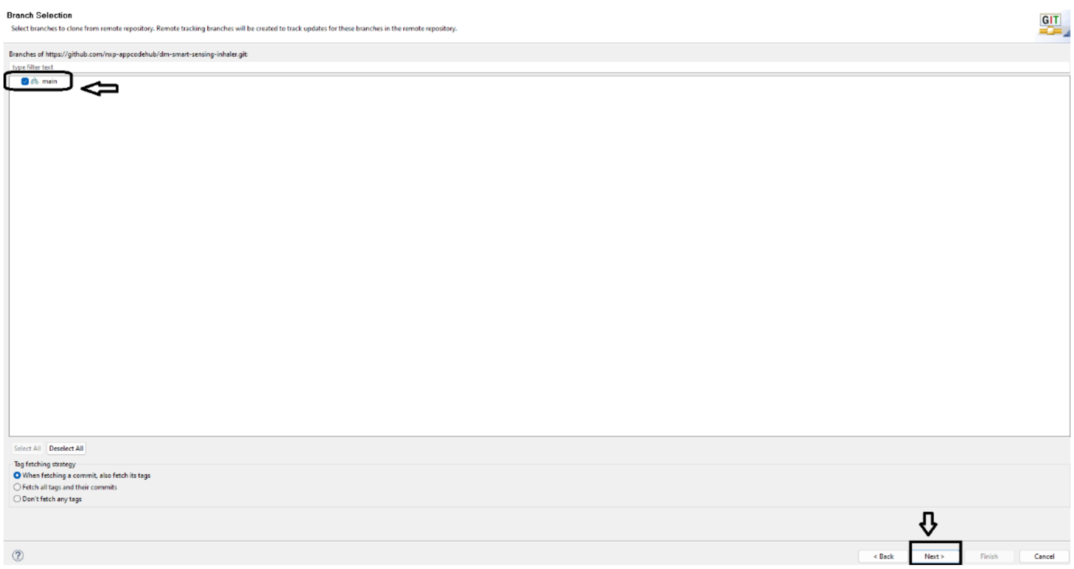
4) Click on "GitHub link", it will start cloning the chosen example project into workspace folder.



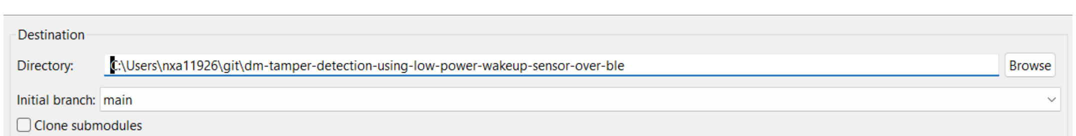
5) The chosen example project is now getting imported into MCUXpresso IDE workspace. Click "Next"



6) Select git branch as “main” and click “Next”.



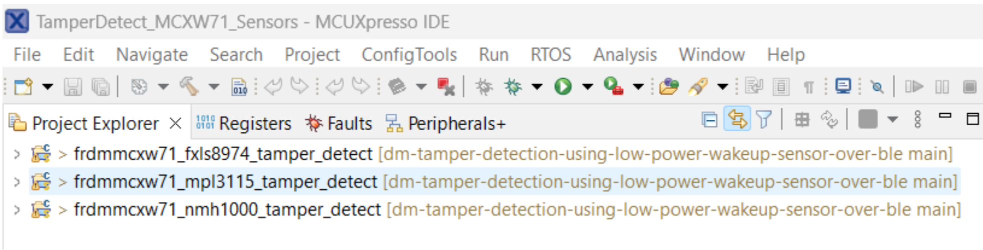
7) Select the local storage locate for cloned project. By default, the tool will select C:\Users\<userid>\git\ folder. Click “Next”.



8) Select projects available under “C:\Users\<userid>\git\dm-tamper-detection-using-low-power-wakeup-sensor-over-ble\tamper_detection_demo”. Click “Next”.

<u>Note</u>: Ignore any “Project SDK management” warning showing older version of SDK. Click on ok button (for all the 3 examples) and continue.

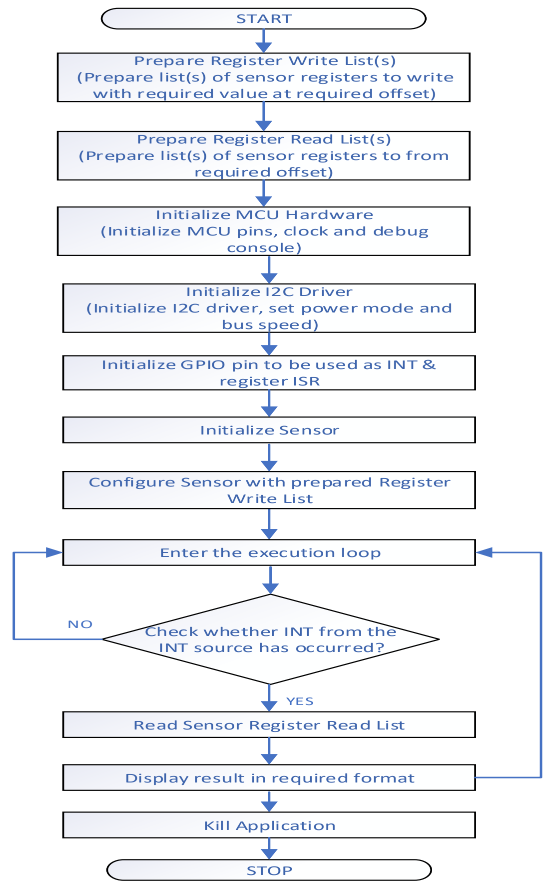
9) Click “Next”. MCUXpresso IDE will open the chosen projects.



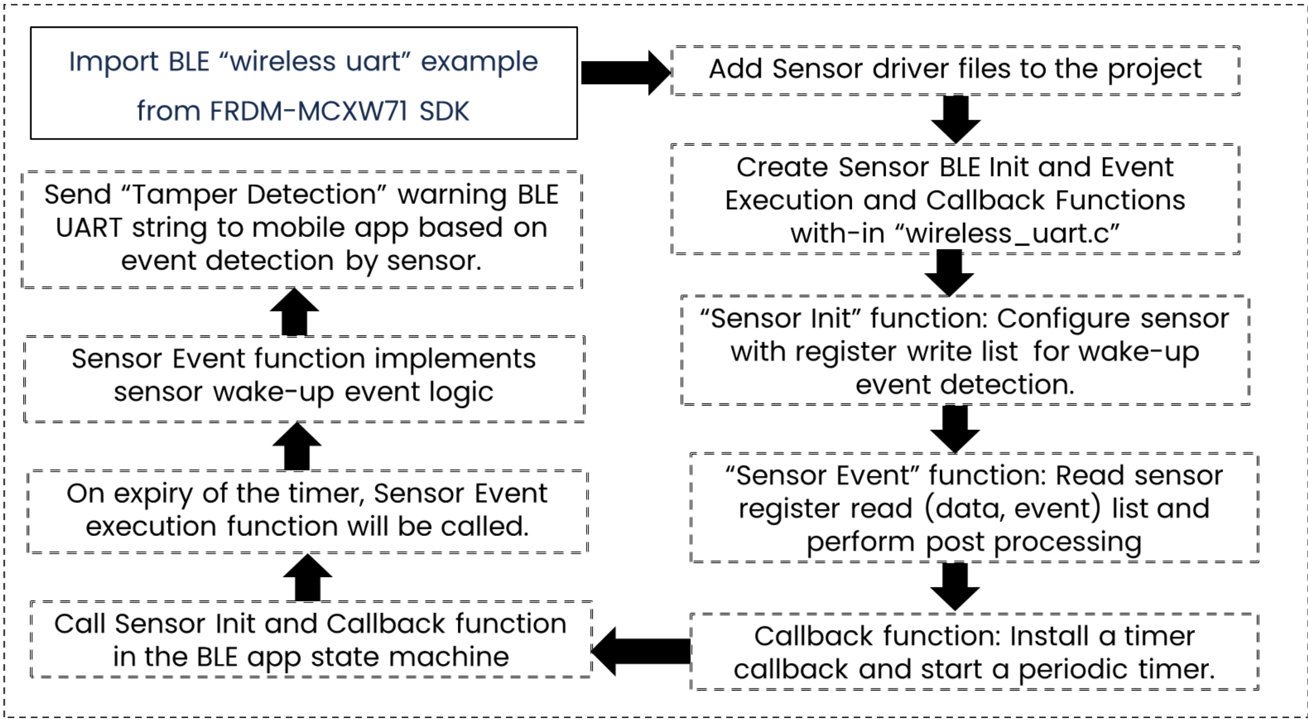
Code Walkthrough

Typical ISSDK example code flow:

1) Prepare register write list.
2) Prepare register read list.
3) Initialize MCU Hardware (Pins, Clock etc.)
4) Initialize COMM (I2C, SPI) driver
5) Initialize GPIO pin to be user as INT and register ISR.
6) Initialize Sensor, read WHOAMI
7) Configure Sensor with prepared register write list
8) Enter execution loop:
   i. Check whether interrupt source has occurred.
   ii. If yes, then read register read list
   iii. Display result or upload into TSA table
   iv. Execute post processing or other functions.

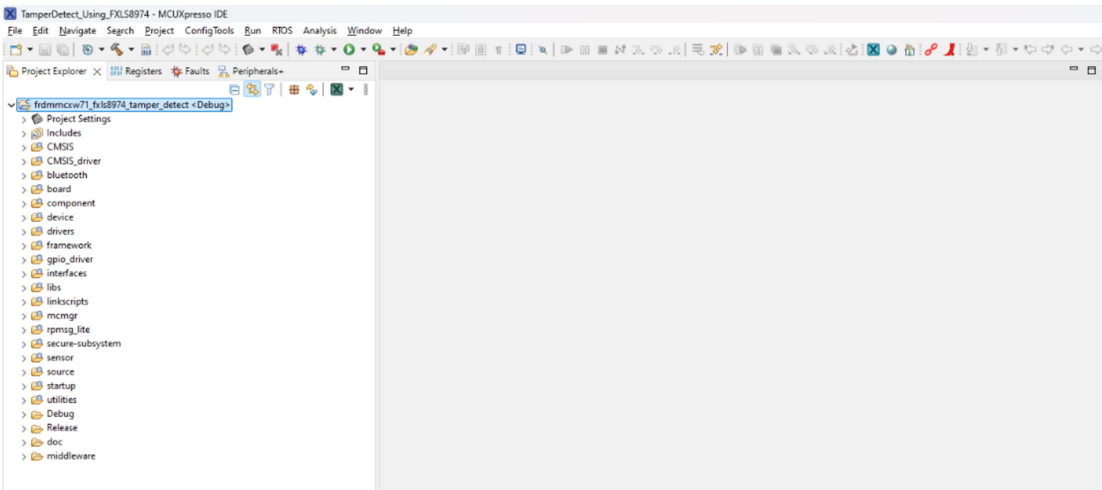


Learn how to convert SDK wireless example to wireless sensing example:



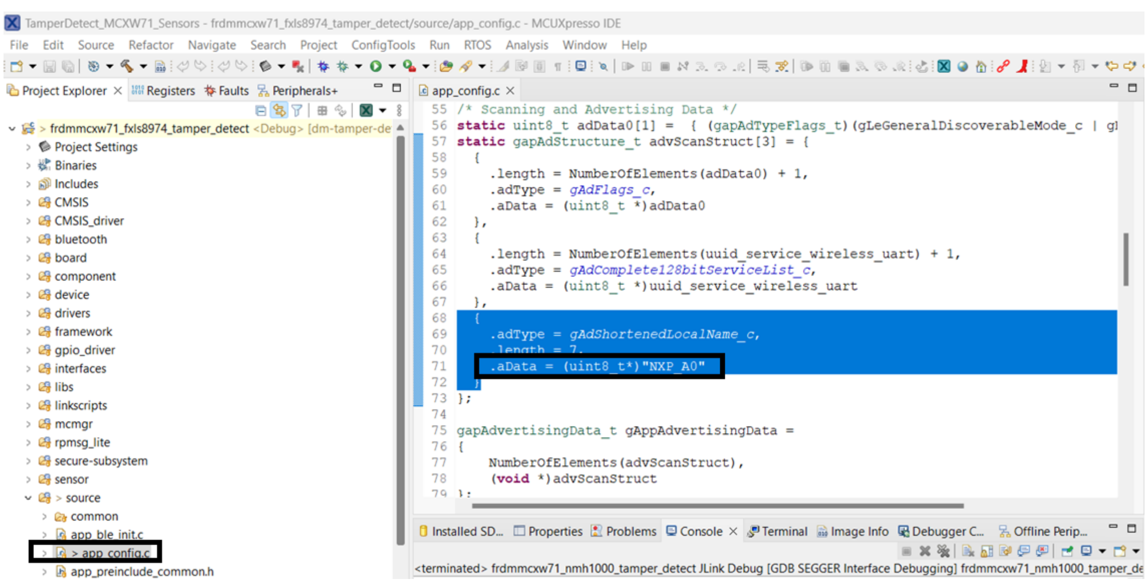
Compile Project:

1) Chose "frdmmcxw71_fxls8974_tamper_detect" project" on your MCUXpresso IDE workspace "TamperDetect_MCXW71_Sensors":

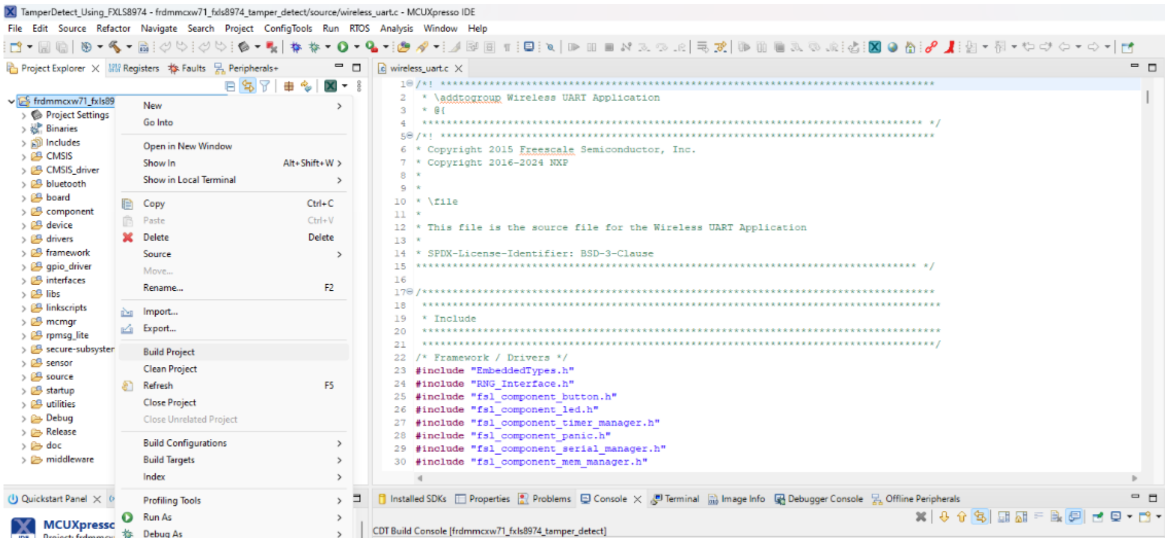


2) Before to compile the project go to Explorer view and open app_config.c file.

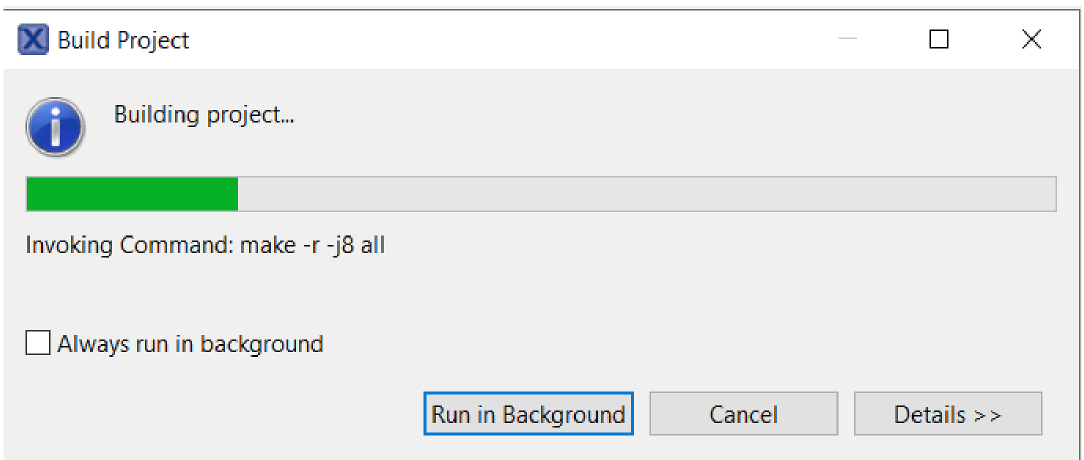
3) On line 71, make sure to update "aData" field with a unique name (this will help to identify your board so that it doesn't clash with any other nearby board). Let me change to "NXP_A0" (try using NXPA01 onwards) but you can select any name you want (max length is 6 characters).



4) Right click on “frdmmcxw71_fxls8974_tamper_detect” project and select “Build Project”

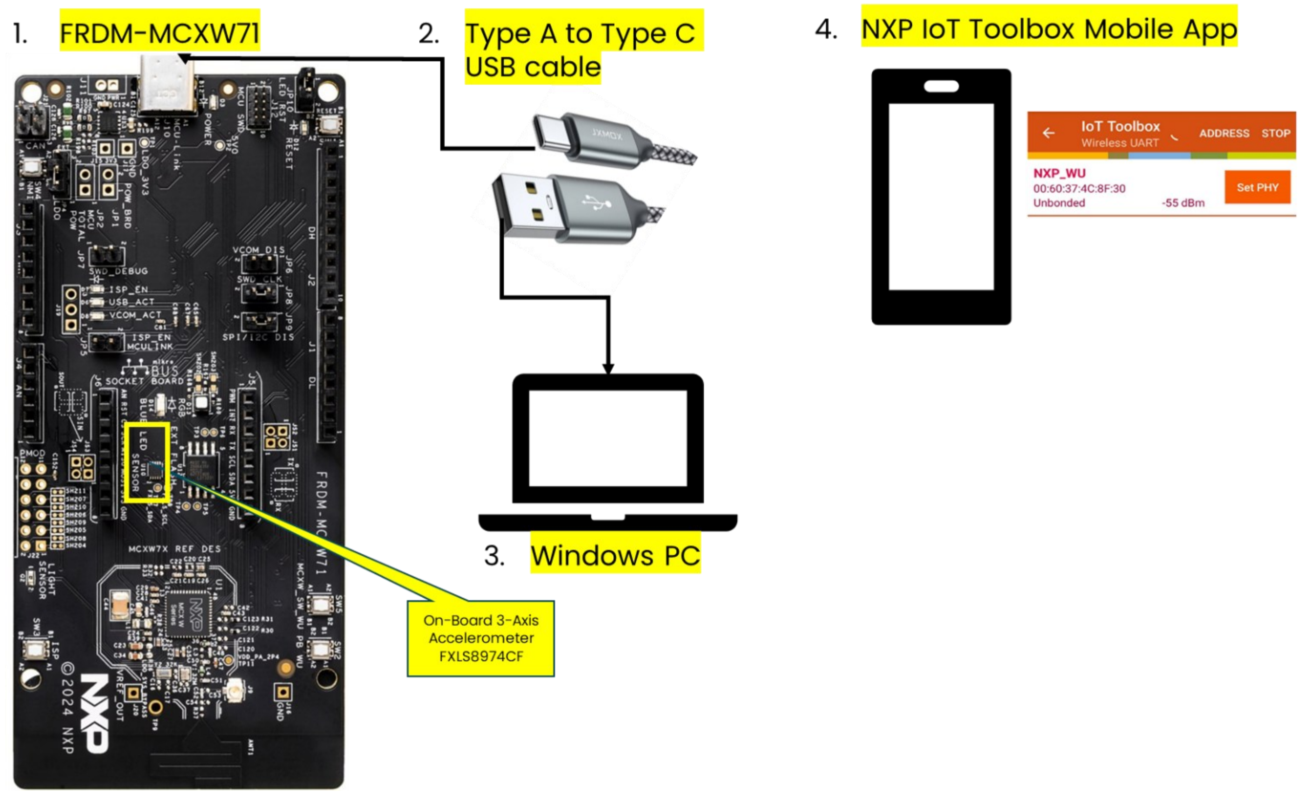


5) MCUXpresso IDE will start compiling the “frdmmcxw71_fxls8974_tamper_detect” project.



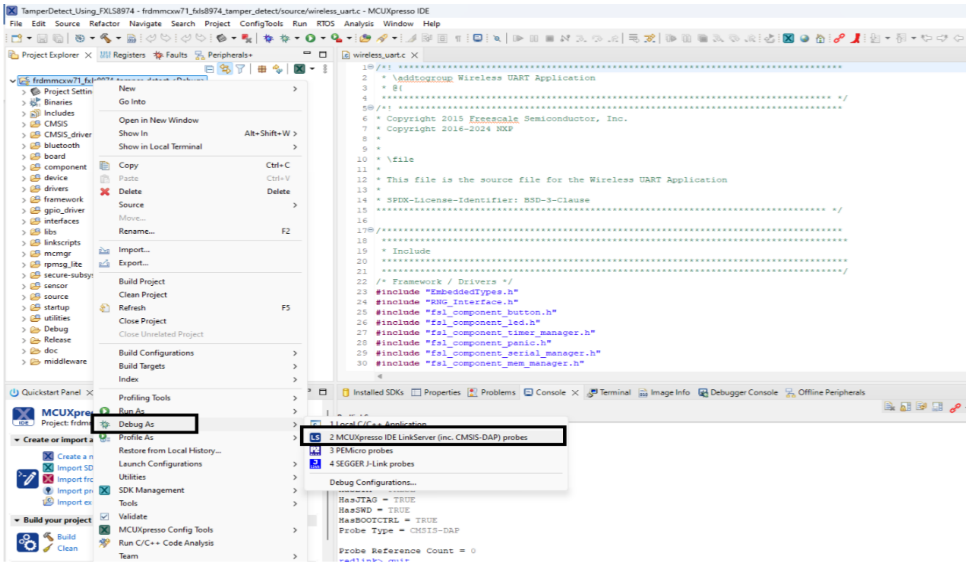
6) Confirm successful compilation of “frdmmcxw71_fxls8974_tamper_detect” project.

Connect HW:

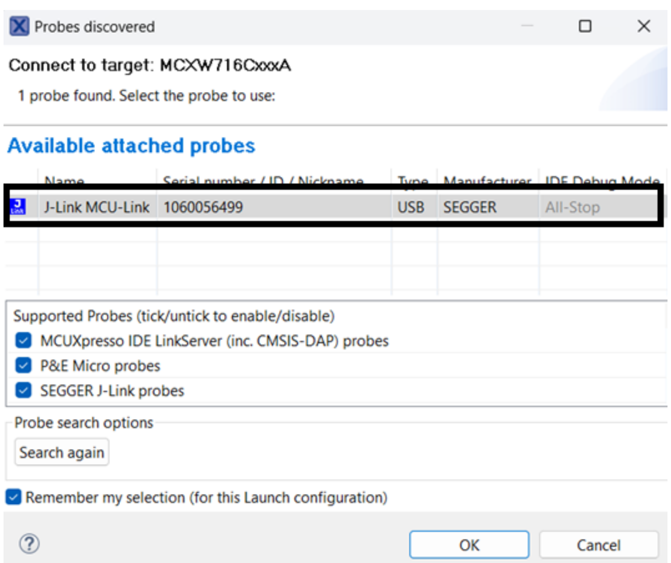

1. FRDM-MCXW71
2. Type A to Type C USB cable
3. Windows PC
4. NXP IoT Toolbox Mobile App
IoT Toolbox
Wireless UART
ADDRESS
STOP
NXP_WU
00:60:37:4C:8F:30
Unbonded
-55 dBm
Set PHY
On-Board 3-Axis Accelerometer FXLS8974CF

Program the target Board:

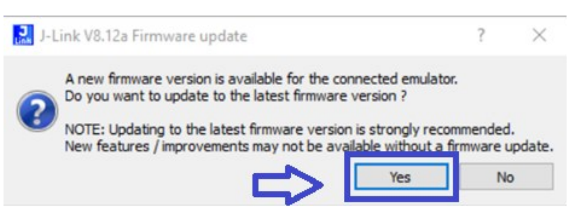
1) Right click on the project and select “Debug As”, click on “SEGGER J-Link probes”.



2) Debugger starts downloading the program to device. “J-Link MCU-LINK” probes will be identified. Click “OK” to continue.

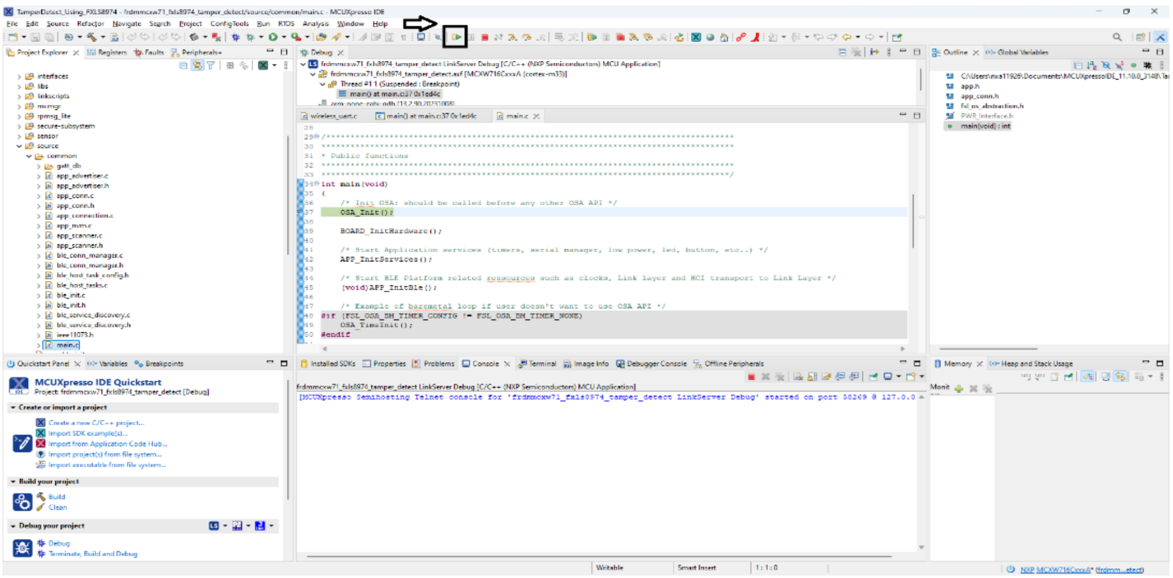


3) In case you see, J-Link FW update pop-up, click on “Yes” to update.



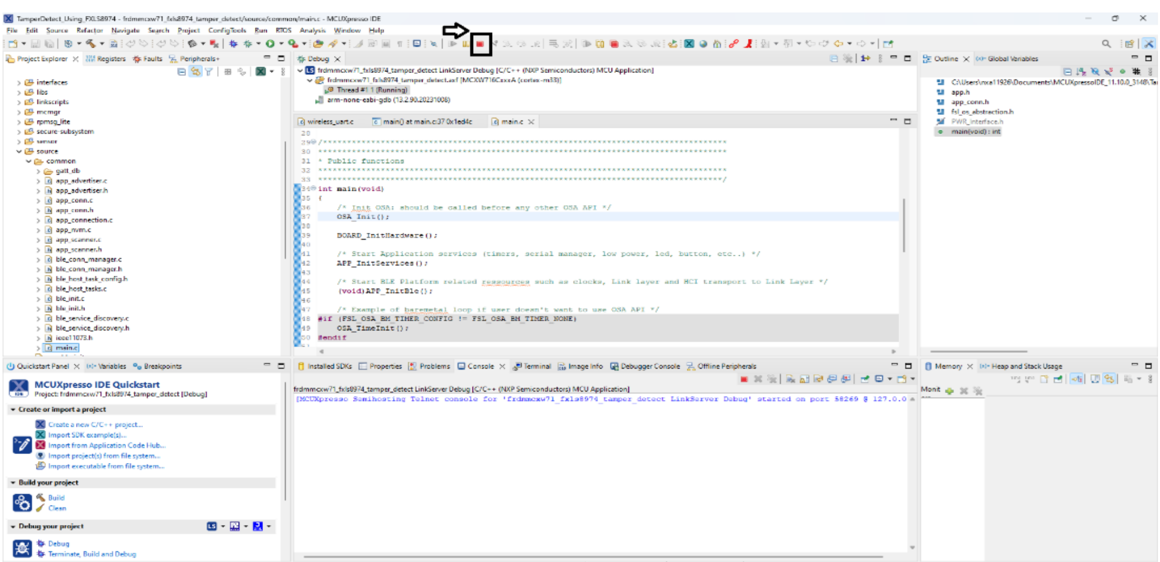
4) Accept the J-Link – Terms of Use. Debugger will load the program to the target board.

5) Click on "Resume" button or press "F8" key on your keyboard to continue running the downloaded program on device.



6) Now the program is downloaded on the device. you should see "BLUE" LED blinking.

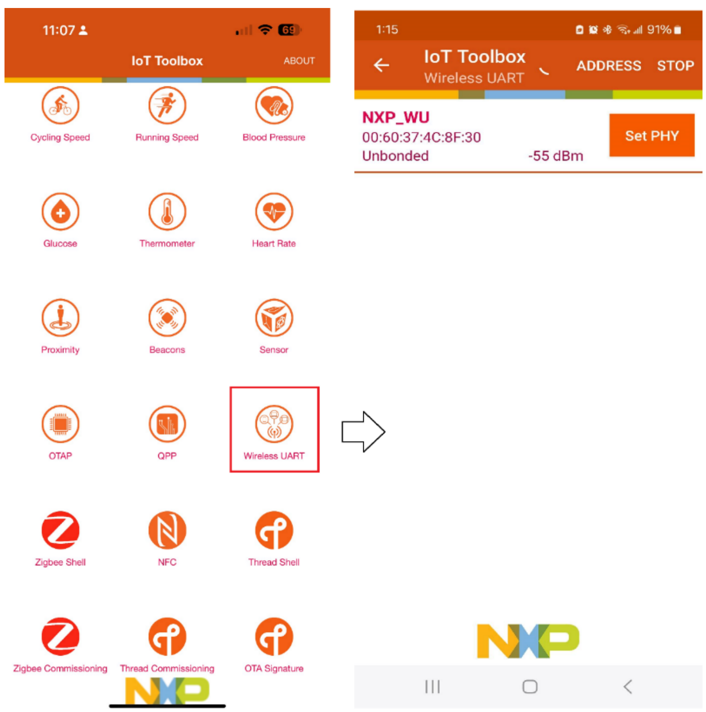
7) Now the program is downloaded on the device. Click on "Terminate" button or press "CNTR + F2" to terminate the debug session.



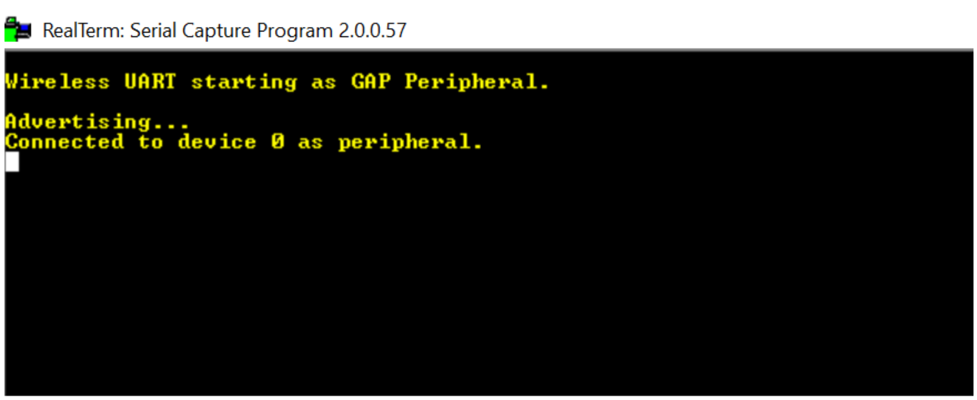
8) Disconnect the USB-C cable connected to FRDM-MCXW71 and reconnect. When powering the FRDM-MCXW71, it starts BLE advertising mode ("BLUE" LED blinking).

Run NXP IoT Toolbox App on your mobile:

1) Open the installed NXP IoT Toolbox (available from Play Store and App Store) on your cell phone. Make sure Bluetooth connection is ON.
2) Inside the application, click on the Wireless UART widget. Scan should start automatically.
3) When a device called NXP_A0 or <your_id> appears, click on it. Your phone should now be connecting to the board.



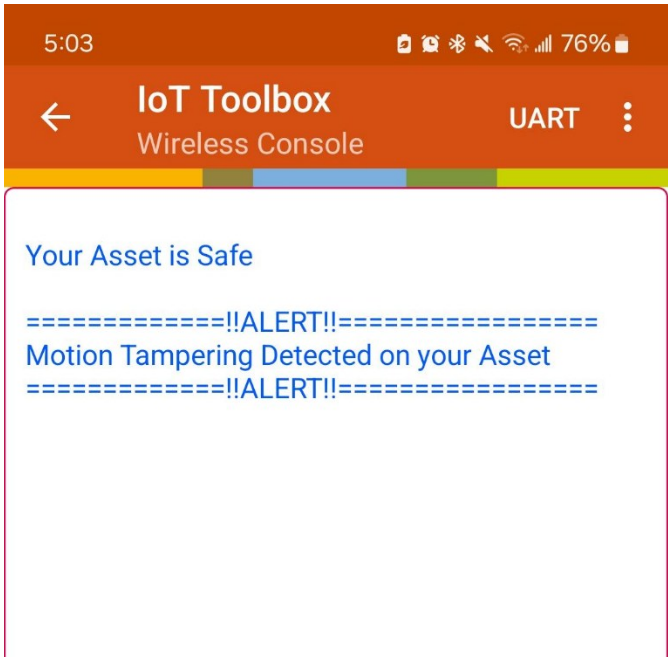
4) After connecting with the FRDM-MCXW71 device, the serial UART terminal will show this message:



5) With no tampering/motion shown on the FRDM-MCXW71 board, the Wireless UART application on IoT ToolBox mobile app will show status as: "Your Asset is Safe“

6) When you show tampering/movement on the FRDM-MCXW71, the on-board FXLS8974CF accelerometer detects the motion and wakeup.

7) At that point, you will see Wireless UART app showing ALERT message as shown below:



8) FRDM-MCXW71 board will also show "RED" LED status. The "RED" LED status will continuously remain ON till tampering/motion detected.

9) If there is no further tampering/motion detected for continuous ~5 sec, the on-board FXLS8974CF accelerometer will detect no-motion and update the status message on wireless UART app. The "RED" LED status on FRDM-MCXW71 board will also go OFF.

Congratulation on successfully completing Hands-On Lab1

Congratulations!

You successfully completed the Hands-On Lab#1